

AMPEL: Adaptive Multi-Phased Embedded Traffic Light System

Moritz Schuster
03771488 IN2060

Jonas Hohenstatter
ge87nes IN2060

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Motivation (Moritz) - smart city - easy to setup, no cable necessary - setup for ampeln for baustelle

System properties/ Framework (Jonas) - Hardware (4 ESPs, ampeln, countdown timer) - Range (gemessen: 15,6m) - C; PlatformIO, ESP-IDF, EspNow, esp-idf-tm1637

Prototype (2 ESP) (*Jonas, Arschloch) - Dedicated Master, Slave - Hardcoded Phases

Final Design: - Design: (Moritz) - Same code on all devices, Master orchestrates Setup, dedicated cmds - Roles - States (Phases) - Communication Diagram - Modularity (Group ID, Session ID, WIFI channels, lights, buttons, CMD) (Code im Detail erklären) (Jonas) - Config (Defines, `ampel_conf_t`, `button_config_t`) - INIT phase; Join, *Time_sync*–*Masteroffset*–*Delaycalculation*(*Probe*, *RTT*)–*RUN*–*Resilience*(*FailedMaster/Slave*; *Switchingthelights*(*CMD*)–*Countdown*(*inMaster*)–*Skippingahead*(*pedestrians*) => *button_pressed*, *interrupt*, *sendnewcommand*

Outlook: (Moritz) - Fussgaenger ampeln - check if light is actually on - Expanding groups - multiple setups (using session ID, don't interfere) - additional commands (apart from switch lights) - configure via build (groups, master, group size) - security (no encryption atm) - risk of jamming signal - Master negotiation - Threads are non deterministic - config over the air/ during build

MÖGLICHERWEISE: modeling, design und analysis approach

1 Introduction

2 System Properties

This section is used to outline and analyze the underlying system properties such as the hardware and software that was used for this project.

2.1 Hardware

The distributed fashion of this project requires the use of multiple microcontrollers with access to wireless communication. To achieve this, a total number of four (ESP 32-WROOM) microcontrollers was chosen as the computational cluster. This 32-bit chip offers 520 kB of RAM and a dual core architecture at 240 MHz. Additionally, it features a Wi-Fi antenna with a bit rate of up to 150 MB s^{-1} as well as a Bluetooth transmitter [?]. For the interaction with the controlled cluster, the allows access to 32 GPIO pins with different capabilities. The layout and design of the used board is visible in Figure ??.

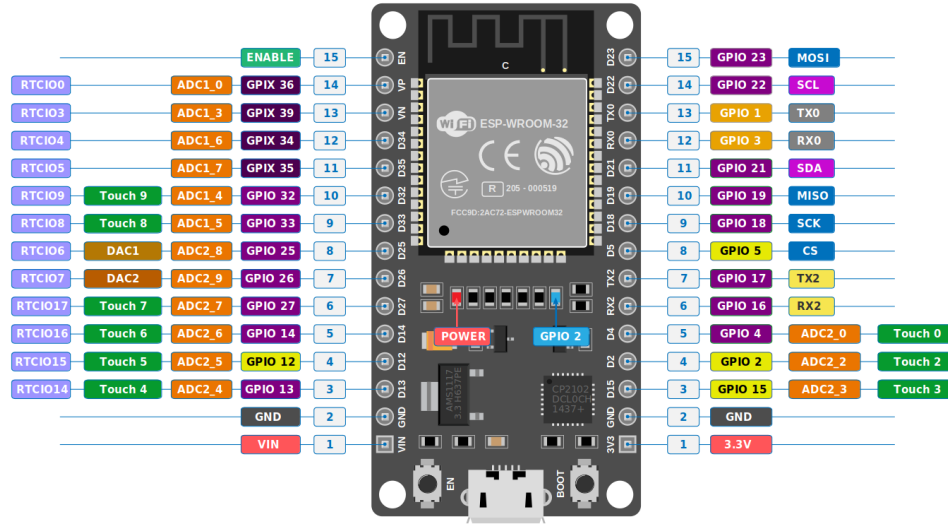


Figure 1: Pinout diagram of the ESP32-WROOM module [?].

To display the current state of each traffic light, multiple prebuilt LED modules were used for simplicity. Each of them features a red, yellow and green LED, that can each be individually controlled using an input at the base of the module. The method of output can be freely chosen and even set up to function with real-world traffic lights. As an additional output the AMPEL can use a 4-digit 7-segment TM1637 display to show numbers and ASCII characters. This external module can be configured using a 4-pin interface that allows for a continuous data flow to the display.

2.2 Software

The program that is running on the Espressif microcontrollers was created on the basis of the ESP-IDF framework using the PlatformIO development platform. This program is compiled and flashed from a C codebase and afterwards executed using the FreeRTOS operating system for real-time systems. FreeRTOS supplies the possibility to use threads and other concurrent execution utilities, which are crucial for the parallel output and

communication capabilities of the AMPEL. It additionally includes timing capabilities that are used to effectively schedule tasks to be executed at a set timestamp.

The communication between the ensemble of microcontrollers is handled by **ESP-Now**, which is a connectionless peer-to-peer protocol for wireless communication over Wifi.

3 Final Design

3.1

Design

3.2

Was weiss ich

References

- [1] Espressif Systems. *ESP32 Series Datasheet*, 2025. v5.2.